



8.2 General Squeeze Film Damper (Standard Bearing)

A general squeeze film damper currently can be created as bearing type “Standard” and imported dimensionless data. The dimensionless table may stem from the inner film of a squeeze film damper. This is a temporary workaround, until a standalone squeeze film damper will be available.

9. Improvements and New Features for Fluid Film Bearings

9.1 General Improvements

The speed and convergence of ALP3T bearing analyses have been considerably improved. This helps for extreme cases such as tilting pad bearing with small preload and central support.

The hydrostatic pocket width can be defined. In previous releases it had a standard width of 98% bearing width. This feature is implemented since version 3.7.4 and described in the current documentation II.6.11

9.2 Closed Cylindrical Bearing

The RFB GUI for the closed bearing is shown in figure 9.1. To model this type of bearing the pad type “Closed” is introduced. In order to supply fresh lubricant to the bearing an axial pressure has to be defined for the closed bearing. Closed bearings do not have any pockets. The pad GUI therefore is simplified. It is shown in figure 9.2.

The analysis types of a closed bearing are restricted to ALP3T_T=c_ad (constant adiabatic) and ALP3T_T=v_ad (variable adiabatic) with 2-phase flow. This is due to the fact, that closed bearings mostly run in high ambient pressure and that cavitation in large areas is suppressed, which requires 2-phase flow for correct modelling.

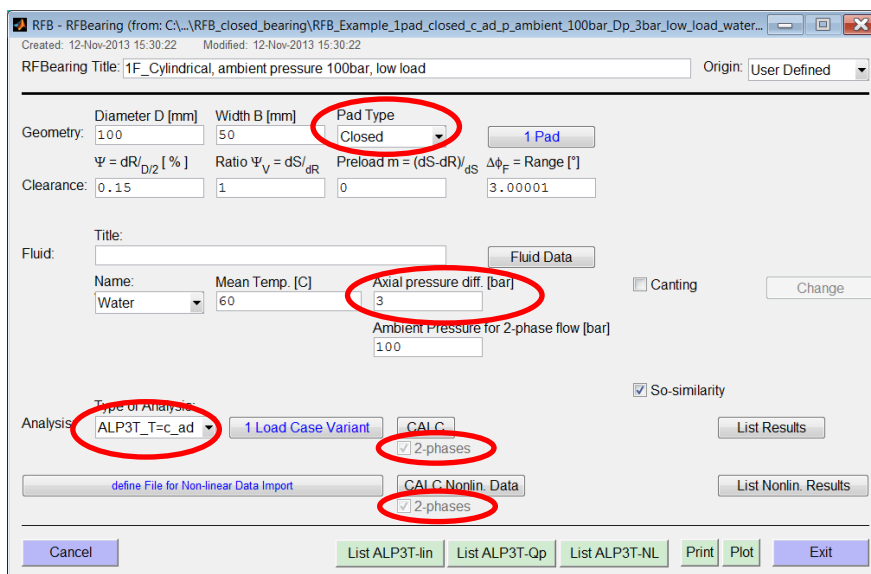


Fig. 9.1: RFB GUI for the closed bearing



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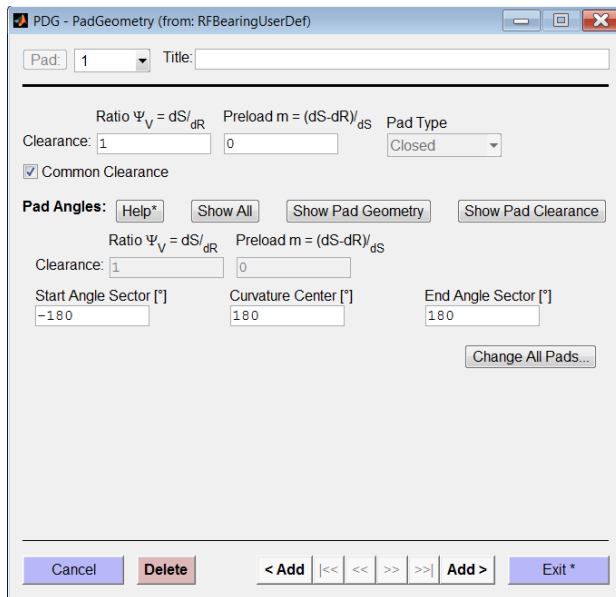


Fig. 9.2: Pad GUI for the closed bearing

The static results and coefficients of a cylindrical water lubricated bearing in high ambient pressure for a rather low load of 100N can be seen in figure 9.3 and 9.4. It can be clearly seen, that the position angle is 90° even at relatively low speed, that the stiffness matrix at elevated speed ($> 50\text{cps}$) is skew symmetric ($k_{32} = -k_{23}$, $k_{22} = k_{33} \approx 0$) and that the damping matrix is diagonal ($d_{23} = d_{32} \approx 0$) and equal in both directions $d_{22} = d_{33}$). These are typical characteristics for a low loaded closed cylindrical bearing.

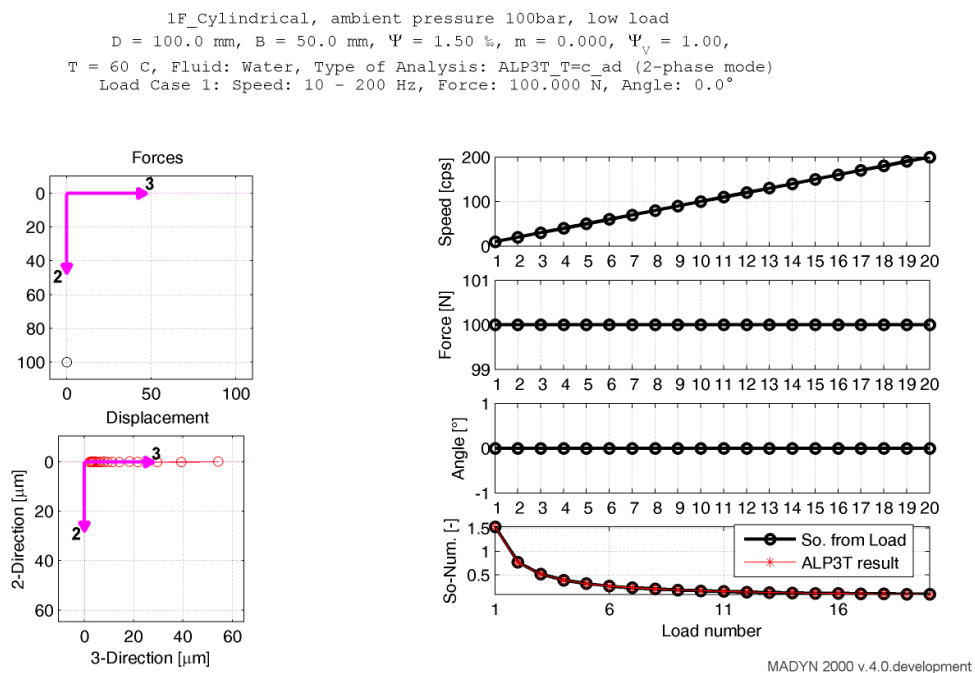


Fig. 9.3: Static results of a closed water lubricated bearing in high ambient pressure (100bar)



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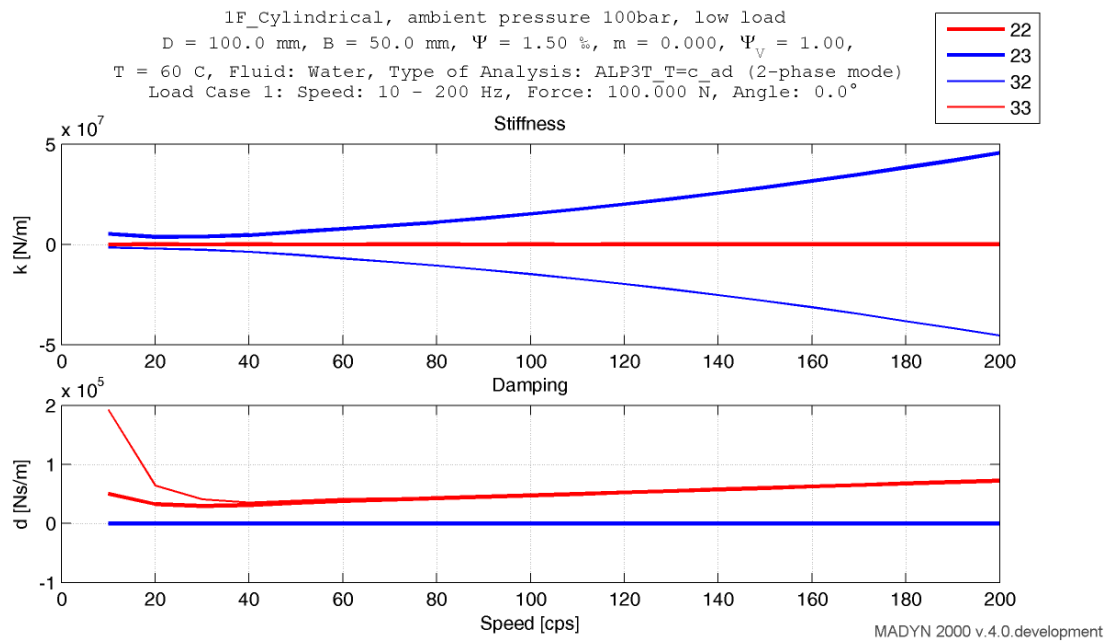


Fig. 9.3: Coefficients of a closed water lubricated bearing in high ambient pressure (100bar)